

DNS AND ROUTE 53 FOR DOMAIN MANAGEMENT

1. Domain Name System (DNS)

- **DNS** (Domain Name System) is like the *phonebook of the internet*. It *translates human-friendly domain names* (like `www.google.com`) into IP addresses (like `142.250.195.68`). that computers use to *communicate* with each other on the *internet*.

➤ Key Concepts of DNS

- **Domain Name:-** A readable address like `example.com`.
- **IP Address:-** A numerical address like **`192.168.1.1`** that computers use.
- **DNS Server :-** A server that stores domain-to-IP mappings and helps resolve names.
- **DNS Resolver:-** A server (usually your ISP's) that finds the correct IP for a domain.
- **DNS Query:-** The request made by a client to get the IP address for a domain.

❖ How DNS Works

- **User enters** a domain name (`www.amazon.com`) in a browser. DNS performs a lookup process through several steps:

- **Local Cache Check:** Your device first checks its local DNS cache to see if it already knows the IP address
- **Recursive Resolver:** If not cached locally, the request goes to a recursive DNS resolver (usually provided by your ISP)
- **Root Nameservers:** The resolver queries one of 13 root nameservers worldwide, which direct it to the appropriate top-level domain (TLD) server
- **TLD Nameservers:** These servers (for `.com`, `.org`, `.net`, etc.) point to the authoritative nameserver for the specific domain
- **Authoritative Nameserver:** This server provides the actual IP address for the requested domain

❖ DNS Record Types

- DNS uses various record types to store different kinds of information:

1. **A Records:** Map domain names to **IPv4 addresses**
 - **Example:** `example.com. IN A 192.0.2.1`
 - **Use case:** Directing a domain to a web server's IPv4 address
2. **AAAA Records:** Map domain names to **IPv6 addresses**
 - **Example:** `example.com. IN AAAA 2001:db8::1`
 - **Use case:** Supporting IPv6 connectivity for a domain
3. **CNAME Records:** Create **aliases** that point to other domain names
 - **Example:** `www.example.com. IN CNAME example.com`
 - **Use case:** Redirecting `www` subdomain to the main domain or to a CDN
4. **MX Records:** Specify **mail servers** for email delivery
 - Includes a priority value to determine the order of mail servers
 - **Example:** `example.com. IN MX 10 mail.example.com`
 - **Use case:** Directing email for a domain to the appropriate mail servers
5. **TXT Records:** Store **text information**, often used for verification
 - **Example:** `example.com. IN TXT "v=spf1 include:_spf.google.com ~all"`
 - **Use cases:** Email validation (SPF), domain ownership verification, security policies
6. **NS Records:** Identify the **nameservers** for a domain
 - **Example:** `example.com. IN NS ns1.example.com`
 - **Use case:** Delegating DNS authority to specific name servers

❖ Types of DNS Servers

- **Root Servers** – Direct the resolver to TLD servers.
- **TLD Servers** – Handle top-level domains like .com, .org, .net.
- **Authoritative Servers** – Hold actual domain-to-IP records.

❖ DNS Zones

- A DNS zone is a portion of the DNS namespace that is managed by a specific organization or administrator.
- It contains DNS records for the domains within that zone.

❖ Types of DNS Zones

1. Public Hosted Zones

- Contain records that specify how to route traffic on the internet
- **Characteristics:** : Publicly accessible, used for internet-facing resources

2. Private Hosted Zones

- Contain records that specify how to route traffic within an Amazon Virtual Private Cloud (VPC)
- **Characteristics:** Not publicly accessible, used for internal resources

3. Forward Zones

- **Used** to forward DNS queries to another DNS server
- **Characteristics:** Delegates resolution to another DNS server
- **Use case:** Forwarding queries for a specific domain to another DNS server

4. Reverse Zones

- Used for reverse DNS lookups (PTR records)
- **Characteristics:** Maps IP addresses to domain names

[GENERAL INTERVIEW QUESTIONS]

❖ Q 1. What is DNS and how does it work?

- DNS (Domain Name System) translates **domain names (like example.com)** into **IP addresses** so that browsers and apps can reach the correct servers.

Q2. What are the main components of a DNS system?

- Domain Names
- Records (A, AAAA, MX, CNAME, etc.)
- Name Servers (NS)
- Resolvers and Authoritative DNS Servers

Q3. What is the difference between an A record and a CNAME?

- **A record** maps a domain to an **IPv4 address**
- **CNAME** maps a domain to **another domain name**

Q4. What is a hosted zone in Route 53?

- A **collection of DNS records** managed under a domain, either **public** (Internet-facing) or **private** (for internal VPC routing).

Q5. What happens when you purchase a domain in Route 53?

- AWS automatically creates a **public hosted zone**, sets up **default name servers**, and gives you full control to add DNS records

Q6. What is TTL and how does it affect DNS changes?

- TTL (Time to Live) defines how long DNS records are **cached by resolvers**. Short TTLs allow **faster updates**, long TTLs reduce DNS traffic.

Q7. Can you route traffic based on user location in Route 53?

- **Yes**, using **Geolocation Routing Policy** — allows region-based content delivery.

Q8. What are NS records, and why are they important?

- NS (Name Server) records define **which name servers are authoritative** for the domain. Without them, DNS resolution fails.

Q9. What's the difference between a Public and Private hosted zone?

- **Public:** Accessible globally over the Internet
- **Private:** Used for internal DNS within a VPC

Q10. Can you integrate Route 53 with Load Balancers?

- **Yes**. Use Alias records to map domain names to ALB or NLB endpoints.

Q11. How can DNS failover be achieved in Route 53?

- Use failover routing with health checks. Route 53 will direct traffic to a secondary endpoint if the primary fails.

Q12. How do you secure DNS in Route 53?

- Restrict changes via **IAM policies**
- Use **MFA**
- Monitor with **CloudTrail**
- Use **Private hosted zones** for internal services

Q13. What are the billing considerations in Route 53?

- Hosted zone charges per **domain/month**
- DNS query costs (millions of queries/month)
- Domain registration is billed separately